

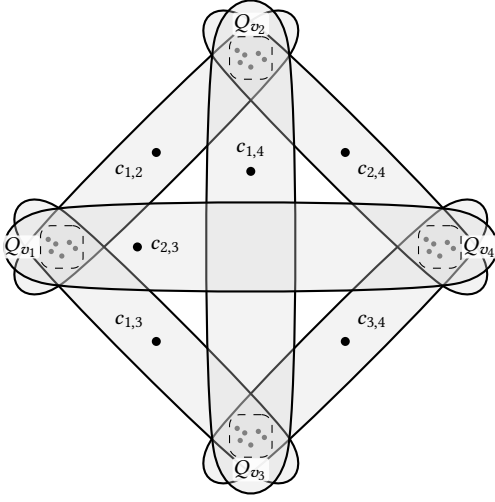


Figure 6: The hypergraph H obtained by the reduction of Theorem 4 on $G = K_4$.

APPENDIX

More experiments

This section of the appendix presents additional plots that complement those shown in Section 7 but were omitted from the main paper for brevity. Figure 7 reports the ratio between the running time and the peak memory usage of the build-up phase of Algorithm 1 and HYPERMOTIVO, as in Section 7.2. Results for 3-hypergraphlets and 4-hypergraphlets are included. Figure 8 shows the same underlying wall-clock times as Figure 7, but plotted as absolute times for each of the two algorithms, side by side, rather than as speedup ratio. Figures 9 and 10 shows more in detail relative frequencies of most frequent 4 and 5-hypergraphlets, respectively. Finally, Figures 11 to 13 illustrate how HYPERMOTIVO scales across different datasets under varying numbers of threads and values of k .

Proof of Theorem 1

This section of the appendix is dedicated to the proof of Theorem 1. In particular, we prove the following statement.

Theorem 4. *There exists an FPT-reduction from κ -CLIQUE to κ -SH that, for every instance (G, k) of κ -CLIQUE, yields an instance (H, k') of κ -SH with the following properties:*

- $k' = \binom{k}{2}(k+1) \leq \frac{k^3}{2}$.
- H is $(k^2, 0)$ -nice.

Moreover, the reduction runs in time $O(k^2 \cdot (|E(G)| + |V(G)|))$.

This clearly implies that κ -SH is $W[1]$ -hard. Theorem 1 follows from Theorem 4 combined with the well-known lower bound $n^{\Omega(k)}$ of κ -CLIQUE under ETH [10]. Note that, by the second item, κ -SH remains $W[1]$ -hard and is subject to a $n^{\Omega(\sqrt[3]{\kappa})}$ ETH-conditional lower bound even if one takes into account the (α, β) -niceness of the hypergraph by adopting as a parameter $\kappa = k + \alpha + \beta$.

The rest of this section gives the proof of Theorem 4.

The reduction. Let (G, k) be an instance of κ -CLIQUE. We construct the corresponding instance (H, k') of κ -SH as follows. For each vertex $v \in V(G)$, create a set $Q_v = \{a_v^1, \dots, a_v^{\binom{k}{2}}\}$, and for each edge $e \in E(G)$, create a singleton $R_e = \{c_e\}$. Define the hypergraph H by

$$V(H) = \bigcup_{v \in V(G)} Q_v \cup \bigcup_{e \in E(G)} R_e \quad (35)$$

and

$$E(H) = \{E_{uv} : e = \{u, v\} \in E(G), E_{uv} = Q_u \cup Q_v \cup R_e\}. \quad (36)$$

Finally, we set $k' := (k+1)\binom{k}{2}$, and we take (H, k') as the output instance of κ -SH.

See Figure 6 for a visual representation of the reduction. Intuitively, one can also think of this construction as encoding a weighted graph G' on the same vertex and edge set as G , where each vertex has weight $\binom{k}{2}$ and each edge weight 1.

We now prove that the construction above is indeed an FPT reduction that satisfies the claim of Theorem 4.

CLAIM 1. *The construction above runs in time $O(k^2 \cdot (|E(G)| + |V(G)|))$.*

PROOF. Let (G, k) be an instance of κ -CLIQUE. For each vertex $v \in V(G)$ we create the set Q_v of size $\binom{k}{2}$, which takes a total of $O(k^2 \cdot |V(G)|)$ time. Then, for each edge $e = \{u, v\} \in E(G)$ we create the vertex c_e and the hyperedge $E_{uv} = Q_u \cup Q_v \cup \{c_e\}$, of size $2\binom{k}{2} + 1 = O(k^2)$, for a total time of $O(k^2 \cdot |E(G)|)$. Summing these two bounds gives the claimed running time $O(k^2 \cdot (|V(G)| + |E(G)|))$. $\square$

The next observation is useful to prove the correctness of the reduction. From now on, $H[U]$ refers to the section hypergraph induced by $U \subseteq V(H)$, as formalized in Definition 2.3.

Observation 1. *Let (G, k) be an instance of κ -CLIQUE and (H, k') the instance of κ -SH obtained via the reduction of Theorem 4. Let $U \subseteq V(H)$ be such that $H[U]$ is connected and $|U| \geq 2$. Then:*

- For every $v \in V(G)$, either $Q_v \cap U = \emptyset$ or $Q_v \subseteq U$.
- For every $e = \{u, v\} \in E(G)$, if $c_e \in U$ then $Q_u \cup Q_v \subseteq U$.

PROOF. For the first item, suppose for a contradiction that there exists $v \in V(G)$ such that $Q_v \cap U \neq \emptyset$ but $Q_v \not\subseteq U$. Take any $x \in Q_v \cap U$ and any $y \in Q_v \setminus U$. Every hyperedge of H containing x is of the form $E_{vw} = Q_v \cup Q_w \cup R_e$, where $e = \{v, w\}$ for some neighbor w of v in G . However, since $y \notin U$, no such hyperedge E_{vw} is entirely contained in U , and therefore no hyperedge containing x appears in $H[U]$. Hence x is an isolated vertex in $H[U]$. As $|U| \geq 2$ and $H[U]$ is connected, this is impossible. Thus either $Q_v \subseteq U$ or $Q_v \cap U = \emptyset$.

For the second item, if $c_e \in U$ but some vertex of Q_u or Q_v is not in U , then the hyperedge $E_{uv} = Q_u \cup Q_v \cup R_e$ is not entirely contained in U . Thus no hyperedge of $H[U]$ contains c_e , and c_e would be isolated in $H[U]$. Again this contradicts the connectivity of $H[U]$ (for $|U| \geq 2$), so we must have $Q_u \cup Q_v \subseteq U$. $\square$

Intuitively, the observation above states that U cannot "partially pick" a Q_v : as soon as U contains one vertex of Q_v , then it must contain the whole block. Similarly, U cannot contain a vertex c_e without also containing the whole blocks Q_u and Q_v .

CLAIM 2. *The construction above is an FPT reduction from κ -CLIQUE to κ -SH.*

PROOF. We prove that (G, k) is a YES-instance of κ -CLIQUE if and only if (H, k') is a YES-instance of κ -SH. Recall that a YES-instance of κ -SH means that there exists $U \subseteq V(H)$ such that $|U| = k'$ and the section hypergraph $H[U]$ is connected.

($\Rightarrow$) From κ -CLIQUE to κ -SH. Assume that G contains a k -clique $K \subseteq V(G)$; define U as

$$U = \bigcup_{e=\{u,v\} \in E(G[K])} E_{uv} = \bigcup_{e=\{u,v\} \in E(G[K])} (Q_u \cup Q_v \cup R_e), \quad (37)$$

where $G[K]$ is the subgraph induced by K . By construction, each vertex $v \in K$ contributes $\binom{k}{2}$ vertices through Q_v and each edge $e = \{u, v\} \in E(G[K])$ contributes exactly one vertex through R_e . Clearly, since K has $\binom{k}{2}$ edges, $|U| = k \binom{k}{2} + \binom{k}{2} = (k+1) \binom{k}{2} = k'$.

We now show that $H[U]$ is connected. For every edge $e = \{u, v\} \in E(G[K])$ we have $E_{uv} \subseteq U$, hence E_{uv} is an edge of $H[U]$. In particular, for every pair of distinct vertices $u, v \in K$ there is a hyperedge E_{uv} contained in $H[U]$ that includes all vertices of Q_u , all vertices of Q_v , and the vertex c_e corresponding to $e = \{u, v\}$; thus, $H[U]$ is connected.

($\Leftarrow$) From κ -SH to κ -CLIQUE. Assume that (H, k') is a YES-instance of κ -SH. Thus, there exists $U \subseteq V(H)$ such that $|U| = k'$ and $H[U]$ is connected. Intuitively, in what follows we decompose U into whole vertex blocks Q_v and edge vertices c_e , and then use counting and connectivity arguments to show that U must correspond to exactly k such blocks that are pairwise adjacent in G , that is, to a k -clique.

First, define T and S as follows

$$T = \{v \in V(G) : Q_v \subseteq U\} \text{ and } S = \{e \in E(G) : c_e \in U\}. \quad (38)$$

Each vertex $v \in T$ contributes all $\binom{k}{2}$ vertices of Q_v to U , and each edge $e \in S$ contributes the vertex c_e . By construction of H we have

$$V(H) = \bigcup_{v \in V(G)} Q_v \cup \bigcup_{e \in E(G)} R_e, \quad (39)$$

so every vertex of U lies either in some Q_v or in some $R_e = \{c_e\}$. Moreover, by Observation 1, if $v \notin T$ then $Q_v \cap U = \emptyset$, and if $c_e \in U$ then $e \in S$. Hence

$$U = \bigcup_{v \in T} Q_v \cup \{c_e : e \in S\}, \quad (40)$$

and we can express the size of U as

$$|U| = |T| \binom{k}{2} + |S|. \quad (41)$$

Call $t = |T|$ and $s = |S|$. Using $|U| = k' = (k+1) \binom{k}{2}$, we obtain

$$(k+1) \binom{k}{2} = t \binom{k}{2} + s \implies s = (k+1-t) \binom{k}{2}. \quad (42)$$

Next, note that Observation 1 also implies that every edge $e = \{u, v\} \in S$ has both endpoints in T : indeed, since $c_e \in U$, the second item of Observation 1 gives $Q_u \cup Q_v \subseteq U$, so $u, v \in T$ by definition of T . Hence $S \subseteq \binom{T}{2}$. Let $G[T]$ be the graph with vertex set T and edge set S . By construction of H , the connected components of

$H[U]$ correspond exactly to the connected components of $G[T]$: contracting each block Q_v to a single vertex v and deleting the vertices $\{c_e\}$ yields $G[T]$, and conversely every edge of $G[T]$ corresponds to a hyperedge E_{uv} in $H[U]$. Since $H[U]$ is connected, the graph $G[T]$ is connected. Therefore,

$$s \geq t - 1, \quad (43)$$

because any connected graph on t vertices has at least $t - 1$ edges. On the other hand, we clearly have

$$s \leq \binom{t}{2}. \quad (44)$$

We now derive constraints on t using eqs. (42) to (44). In particular, we show that $t = k$, meaning that $s = \binom{k}{2}$ and thus G contains a k -clique. We may assume $k \geq 3$, since for $k \leq 2$ both κ -CLIQUE and κ -SH are trivial.

Case 1: $t \geq k + 1$. From eq. (42) we obtain $s = (k+1-t) \binom{k}{2} \leq 0$. However, $H[U]$ is connected and $|U| = k' > 1$, so $H[U]$ must contain at least one hyperedge, which implies $s \geq 1$. This is a contradiction. Thus $t \leq k$.

Case 2: $t \leq k - 1$. From eq. (42) we have $k+1-t \geq 2$, and hence

$$s = (k+1-t) \binom{k}{2} \geq 2 \binom{k}{2} = k(k-1). \quad (45)$$

On the other hand, since $s \leq \binom{t}{2}$ and $t \leq k - 1$, we have

$$k(k-1) \leq s \leq \binom{k-1}{2} = \frac{(k-1)(k-2)}{2}. \quad (46)$$

But clearly, for $k \geq 3$ we obtain

$$k(k-1) > \frac{(k-1)(k-2)}{2}, \quad (47)$$

which leads to a contradiction. Therefore $t \not\leq k - 1$. The only remaining possibility is $t = k$. Finally, substituting into (42) we get

$$s = (k+1-k) \binom{k}{2} = \binom{k}{2}. \quad (48)$$

Since $s = |S|$ and $S \subseteq \binom{T}{2}$ with $|T| = k$, this means that S contains all possible $\binom{k}{2}$ edges on the vertex set T . Therefore $G[T]$ is a clique on k vertices, and hence G contains a k -clique. This is sufficient to prove the claim. $\square$

CLAIM 3. *The construction above ensures that $k' = (k+1) \binom{k}{2} \leq \frac{k^3}{2}$ and that H is $(k^2, 0)$ -nice.*

PROOF. First point follows simply by definition of the reduction; indeed we have $k' = (k+1) \binom{k}{2} \leq \frac{k^2 \cdot k}{2} = \frac{k^3}{2}$ for $k \geq 1$.

We now show that H is $(k^2, 0)$ -nice. By construction, every hyperedge of H is of the form

$$E_{uv} = Q_u \cup Q_v \cup R_e, \quad (49)$$

where $e = \{u, v\} \in E(G)$. Thus

$$|E_{uv}| = |Q_u| + |Q_v| + |R_e| = \binom{k}{2} + \binom{k}{2} + 1 = k(k-1) + 1, \quad (50)$$

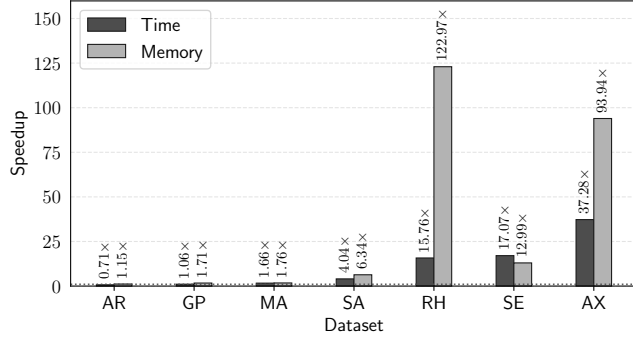
and hence

$$k(k-1) + 1 = k^2 - k + 1 \leq k^2 \quad (51)$$

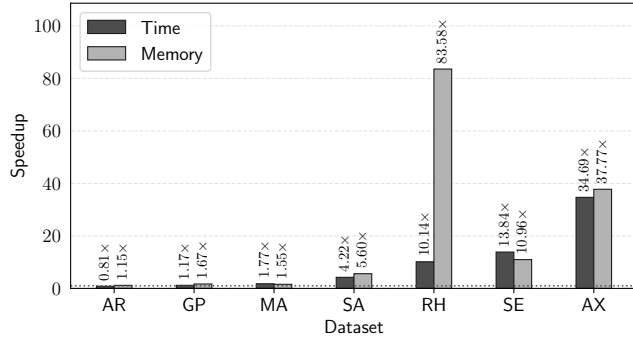
for every $k \geq 1$. Therefore the rank of H (the maximum size of a hyperedge) is at most k^2 .

Consider the α -split of H with $\alpha := k^2$. Since no hyperedge has size greater than α , the upper part $H_{>\alpha}$ contains no hyperedges,

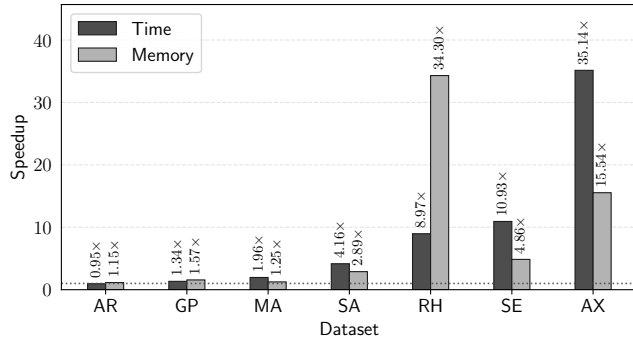
and in particular every vertex has degree 0 in $H_{>\alpha}$. By the definition of (α, β) -niceness, this means that H is $(k^2, 0)$ -nice. $\square$



(a)

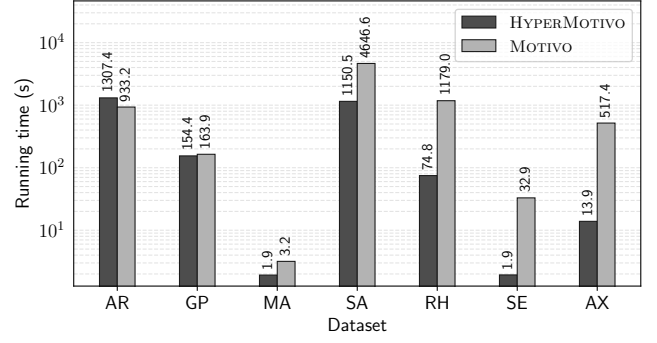


(b)

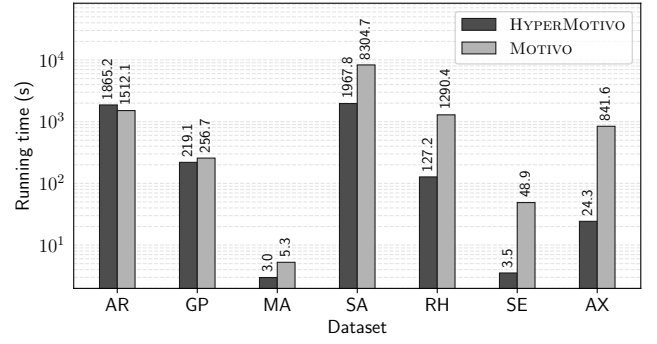


(c)

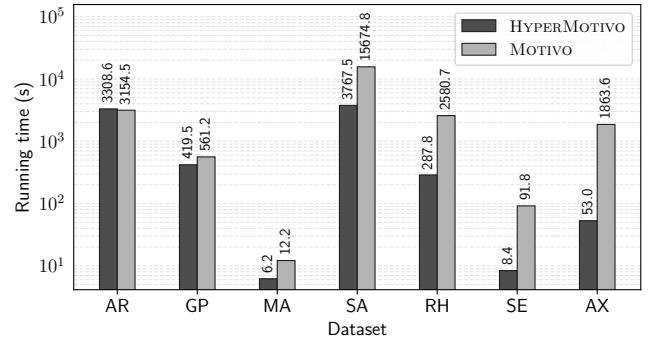
Figure 7: Speedup in wall-clock time and peak memory usage of HYPERMOTIVO over Algorithm 1, on the build-up phase, across all hypergraphs, for (a) 3-hypergraphlets, (b) 4-hypergraphlets and (c) 5-hypergraphlets.



(a)

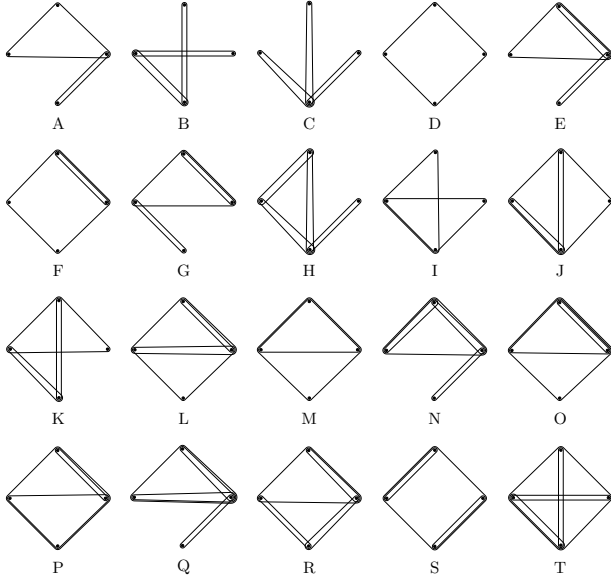


(b)



(c)

Figure 8: Wall-clock time of HYPERMOTIVO and Algorithm 1, on the build-up phase, across all hypergraphs, for (a) 3-hypergraphlets, (b) 4-hypergraphlets and (c) 5-hypergraphlets.

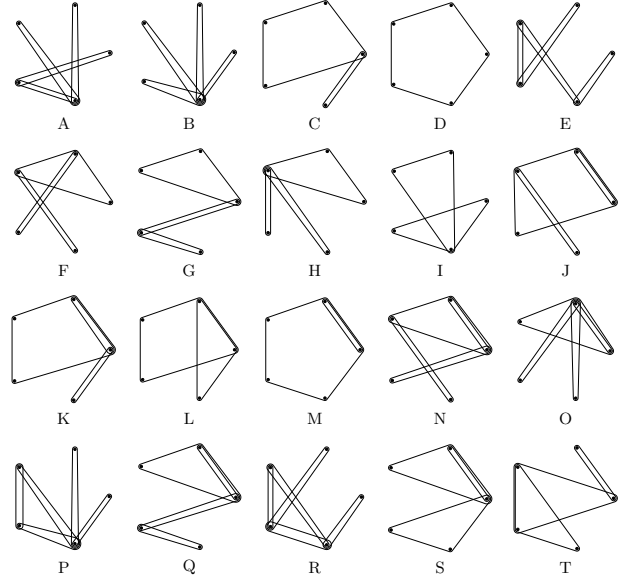


(a)

	AR	AX	GP	SE	MA	RH	SA
A	0.0175	0.3605	0.1136	0.3281	0.3258	0.0692	0.2936
B	0.3649	0.1506	0.3564	0.1851	0.2424	0.0261	0.0461
C	0.5316	0.0450	0.2009	0.0592	0.3043	0.0073	0.0199
D	0.0004	0.0678	0.0107	0.0808	0.0814	0.0323	0.6324
E	0.0105	0.0806	0.0804	0.0790	0.0214	0.0575	0.0010
F	0.0002	0.0672	0.0127	0.0649	0.0042	0.1013	0.0050
G	0.0025	0.0515	0.0360	0.0461	0.0037	0.0287	0.0008
H	0.0543	0.0054	0.0487	0.0148	0.0082	0.0100	0.0001
I	0.0001	0.0515	0.0036	0.0280	0.0023	0.0300	0.0009
J	0.0001	0.0127	0.0060	0.0131	0.0002	0.0817	0.0000
K	0.0008	0.0230	0.0094	0.0288	0.0023	0.0298	0.0002
L	0.0001	0.0223	0.0052	0.0131	0.0002	0.0417	0.0000
M	0.0000	0.0173	0.0011	0.0061	0.0000	0.0442	0.0000
N	0.0029	0.0053	0.0297	0.0086	0.0006	0.0192	0.0000
O	0.0000	0.0049	0.0018	0.0029	0.0000	0.0426	0.0000
P	0.0000	0.0045	0.0010	0.0019	0.0000	0.0428	0.0000
Q	0.0029	0.0027	0.0204	0.0031	0.0010	0.0087	0.0000
R	0.0004	0.0025	0.0061	0.0064	0.0002	0.0202	0.0000
S	0.0000	0.0051	0.0015	0.0041	0.0000	0.0211	0.0000
T	0.0000	0.0008	0.0016	0.0012	0.0000	0.0266	0.0000

(b)

Figure 9: First 20 4-hypergraphlets with highest aggregate frequency across all datasets. Figure 9a presents a visual representation of these hypergraphlets, sorted by descending aggregate frequency. Figure 9b shows in more detail relative frequencies of each hypergraphlet in every hypergraph.



(a)

	AR	AX	GP	SE	MA	RH	SA
A	0.3549	0.0261	0.2148	0.0455	0.2222	0.0015	0.0109
B	0.3941	0.0030	0.0615	0.0040	0.2310	0.0000	0.0015
C	0.0004	0.1070	0.0166	0.1118	0.0667	0.0139	0.2747
D	0.0000	0.0144	0.0013	0.0197	0.0134	0.0057	0.5330
E	0.0908	0.0259	0.1360	0.0533	0.0584	0.0023	0.0095
F	0.0042	0.1055	0.0430	0.1071	0.0715	0.0045	0.0389
G	0.0052	0.0884	0.0425	0.0888	0.0872	0.0048	0.0430
H	0.0080	0.0447	0.0337	0.0465	0.1218	0.0025	0.0206
I	0.0001	0.0640	0.0029	0.0366	0.0303	0.0028	0.0541
J	0.0001	0.0418	0.0083	0.0402	0.0016	0.0159	0.0010
K	0.0003	0.0315	0.0093	0.0328	0.0043	0.0158	0.0016
L	0.0000	0.0419	0.0010	0.0210	0.0013	0.0131	0.0021
M	0.0000	0.0207	0.0019	0.0241	0.0010	0.0244	0.0069
N	0.0022	0.0185	0.0274	0.0205	0.0042	0.0033	0.0000
O	0.0074	0.0060	0.0276	0.0078	0.0239	0.0013	0.0002
P	0.0452	0.0004	0.0170	0.0012	0.0097	0.0000	0.0000
Q	0.0026	0.0110	0.0274	0.0198	0.0055	0.0045	0.0002
R	0.0305	0.0004	0.0181	0.0028	0.0062	0.0000	0.0000
S	0.0001	0.0266	0.0041	0.0172	0.0028	0.0032	0.0004
T	0.0000	0.0294	0.0026	0.0145	0.0011	0.0041	0.0004

(b)

Figure 10: First 20 5-hypergraphlets with highest aggregate frequency across all datasets. Figure 10a presents a visual representation of these hypergraphlets, sorted by descending aggregate frequency. Figure 10b shows in more detail relative frequencies of each hypergraphlet in every hypergraph.

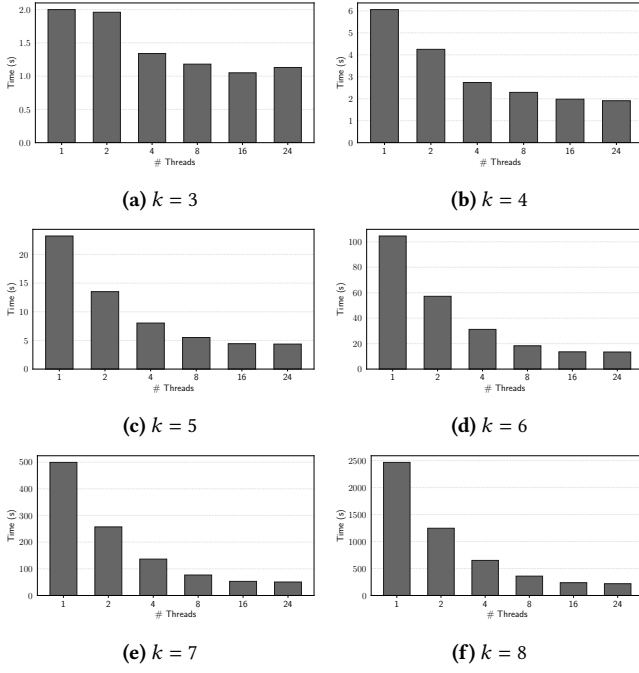


Figure 11: Running time of HYPERMOTIVO's build-up phase on the MA dataset for hypergraphlets of size $k \in \{3, 4, 5, 6, 7, 8\}$ under different thread configurations.

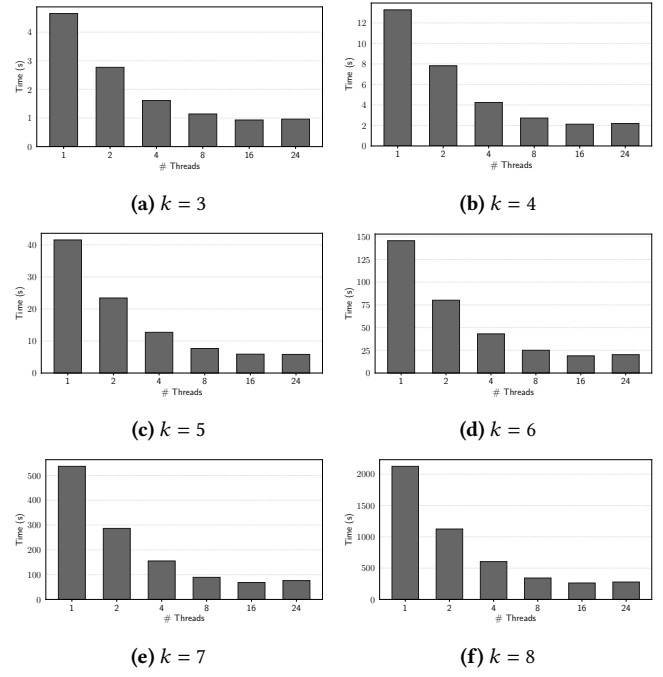


Figure 12: Running time of HYPERMOTIVO's build-up phase on the SE dataset for hypergraphlets of size $k \in \{3, 4, 5, 6, 7, 8\}$ under different thread configurations.

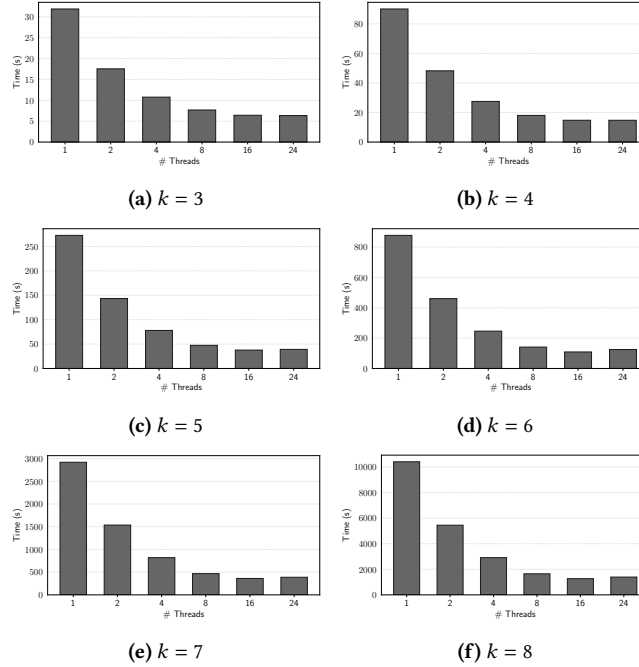


Figure 13: Running time of HYPERMOTIVO's build-up phase on the AX dataset for hypergraphlets of size $k \in \{3, 4, 5, 6, 7, 8\}$ under different thread configurations.